

PRE-MEDICAL : ENTHUSIAST COURSE (ALL PHASE)

This Booklet contains 16 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.



Important Instructions :

1. On the Answer Sheet and fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

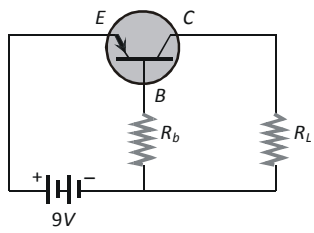
: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

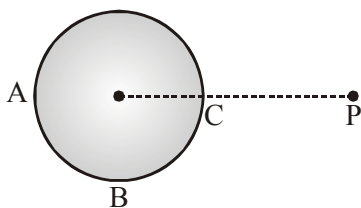
Your Target is to secure Good Rank in Pre-Medical 2020

1. The condition for a uniform spherical mass m of radius r to be a black hole is [G = gravitational constant and g = acceleration due to gravity]
 - (1) $(2Gm/r)^{1/2} \leq c$
 - (2) $(2Gm/r)^{1/2} = c$
 - (3) $(2Gm/r)^{1/2} \geq c$
 - (4) $(gm/r)^{1/2} \geq c$
2. The scale of a spring balance reading from 0 to 10 kg is 0.25 m long. A body suspended from the balance oscillates vertically with a period of $\pi/10$ second. The mass suspended is (neglect the mass of the spring)
 - (1) 10 kg
 - (2) 0.98 kg
 - (3) 5 kg
 - (4) 20 kg
3. A spherical ball of mass 20 kg is stationary at the top of a hill of height 100 m. It slides down a smooth surface to the ground, then climbs up another hill of height 30 m and finally slides down to a horizontal base at a height of 20 m above the ground. The velocity attained by the ball is
 - (1) 10 m/s
 - (2) $10\sqrt{30}$ m/s
 - (3) 40 m/s
 - (4) 20 m/s
4. A current i is flowing in a straight conductor of length L . The magnetic induction at a point distant $\frac{L}{4}$ from its centre will be
 - (1) $\frac{4\mu_0 i}{\sqrt{5}\pi L}$
 - (2) $\frac{\mu_0 i}{2\pi L}$
 - (3) $\frac{\mu_0 i}{\sqrt{2}L}$
 - (4) Zero
5. Two radioactive materials X_1 and X_2 have decay constants 10λ and λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be $1/e$ after a time
 - (1) $1/(10\lambda)$
 - (2) $1/(11\lambda)$
 - (3) $11/(10\lambda)$
 - (4) $1/(9\lambda)$
6. A body has a weight 90 kg on the earth's surface, the mass of the moon is $1/9$ that of the earth's mass and its radius is $1/2$ that of the earth's radius. On the moon the weight of the body is
 - (1) 45 kg
 - (2) 202.5 kg
 - (3) 90 kg
 - (4) 40 kg
7. The echo of a gun shot is heard 8 sec. after the gun is fired. How far from him is the surface that reflects the sound (velocity of sound in air = 350 m/s)
 - (1) 1400 m
 - (2) 2800 m
 - (3) 700 m
 - (4) 350 m
8. A bullet of mass m moving with velocity v strikes a block of mass M at rest and gets embedded into it. The kinetic energy of the composite block will be
 - (1) $\frac{1}{2}mv^2 \times \frac{m}{(m+M)}$
 - (2) $\frac{1}{2}mv^2 \times \frac{M}{(m+M)}$
 - (3) $\frac{1}{2}mv^2 \times \frac{(M+m)}{M}$
 - (4) $\frac{1}{2}Mv^2 \times \frac{m}{(m+M)}$
9. A particle of charge q and mass m moves in a circular orbit of radius r with angular speed ω . The ratio of the magnitude of its magnetic moment to that of its angular momentum depends on
 - (1) ω and q
 - (2) ω , q and m
 - (3) q and m
 - (4) ω and m
10. In a transistor circuit shown here the base current is 35 μ A. The value of the resistor R_b is

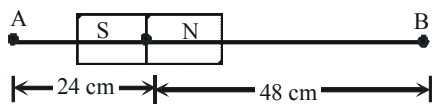


 - (1) 123.5 k Ω
 - (2) 257 k Ω
 - (3) 380.05 k Ω
 - (4) None of these

11. The increase in length on stretching a wire is 0.05%. If its Poisson's ratio is 0.4, then its diameter
- Reduce by 0.02%
 - Reduce by 0.1%
 - Increase by 0.02%
 - Decrease by 0.4%
12. A pipe 30 cm long is open at both ends. Which harmonic mode of the pipe is resonantly excited by a 1.1 kHz source ?
(Take speed of sound in air = 330 ms⁻¹)
- First
 - Second
 - Third
 - Fourth
13. A hollow conducting sphere is placed in an electric field produced by a point charge placed at P as shown in figure. Let V_A, V_B, V_C be the potentials at points A, B and C respectively. Then

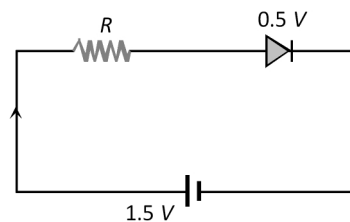


- $V_C > V_B$
 - $V_B > V_C$
 - $V_A > V_B$
 - $V_A = V_C$
14. A bar magnet of length 3 cm has points A and B along its axis at distances of 24 cm and 48 cm on the opposite sides. Ratio of magnetic fields at these points will be

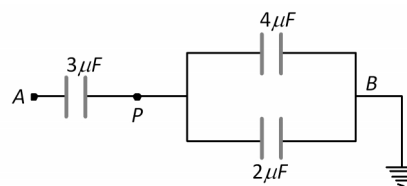


- 8
- $1/2\sqrt{2}$
- 3
- 4

15. The diode used in the circuit shown in the figure has a constant voltage drop of 0.5 V at all currents and a maximum power rating of 100 milliwatts. What should be the value of the resistor R, connected in series with the diode for obtaining maximum current



- 1.5 Ω
 - 5 Ω
 - 6.67 Ω
 - 200 Ω
16. A soap film of surface tension $3 \times 10^{-2} \text{ Nm}^{-1}$ formed in rectangular frame, can support a straw. The length of the film is 10 cm. Mass of the straw the film can support is
- 0.06 gm
 - 0.6 gm
 - 6 gm
 - 60 gm
17. In a football championship, 153 matches were played. Every team played one match with each other. The number of teams participating in the championship is :-
- 17
 - 18
 - 9
 - None of these
18. In the figure a potential of + 1200 V is given to point A and point B is earthed, what is the potential at the point P :-

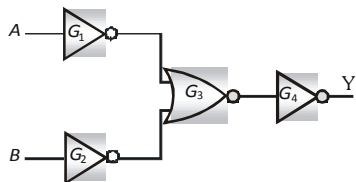


- 100 V
- 200 V
- 400 V
- 600 V

19. In a transformer, the number of turns in primary and secondary are 500 and 2000 respectively. If current in primary is 48 A, the current in the secondary is

(1) 12 A (2) 24 A
(3) 48 A (4) 144 A

20. The combination of gates shown below produces

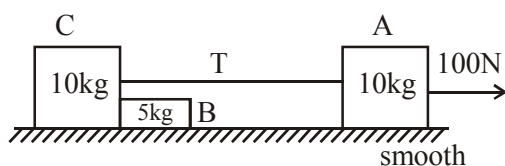


(1) AND gate (2) XOR gate
(3) NOR gate (4) NAND gate

21. One mole of an ideal gas at an initial temperature of T K does $6R$ joules of work adiabatically. If the ratio of specific heats of this gas at constant pressure and at constant volume is $5/3$, the final temperature of gas will be

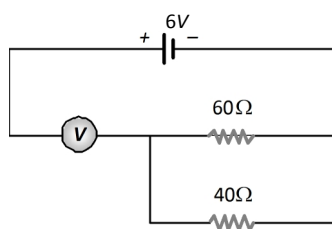
(1) $(T + 2.4)K$ (2) $(T - 2.4)K$
(3) $(T + 4)K$ (4) $(T - 4)K$

22. Contact force between block B & C is :-



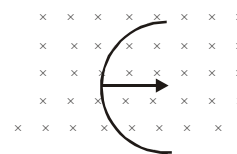
(1) 10N (2) 20N
(3) 40N (4) 50N

23. The measurement of voltmeter in the following circuit is :-



(1) 2.4 V (2) 3.4 V
(3) 4.0 V (4) 6.0 V

24. A straight wire of length L is bent into a semicircle. It is moved in a uniform magnetic field with speed v with diameter perpendicular to the field. The induced emf between the ends of the wire is



(1) BLv (2) $2BLv$
(3) $2\pi BLv$ (4) $\frac{2BvL}{\pi}$

25. A concave mirror and a converging lens (glass with $\mu = 1.5$) both have a focal length of 3 cm

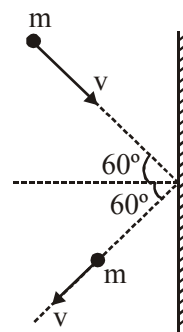
when in air. When they are in water ($\mu = \frac{4}{3}$), their new focal lengths are

(1) $f_{\text{Lens}} = 12$ cm, $f_{\text{Mirror}} = 3$ cm
(2) $f_{\text{Lens}} = 3$ cm, $f_{\text{Mirror}} = 12$ cm
(3) $f_{\text{Lens}} = 3$ cm, $f_{\text{Mirror}} = 3$ cm
(4) $f_{\text{Lens}} = 12$ cm, $f_{\text{Mirror}} = 12$ cm

26. The temperature of reservoir of Carnot's engine operating with an efficiency of 70% is 1000K. The temperature of its sink is

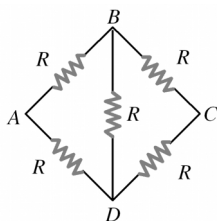
(1) 300 K (2) 400 K (3) 500 K (4) 700 K

27. A rigid ball of mass m strikes a rigid wall at 60° with normal and gets reflected without loss of speed as shown in the figure. The value of impulse imparted by the wall on the ball will be:-



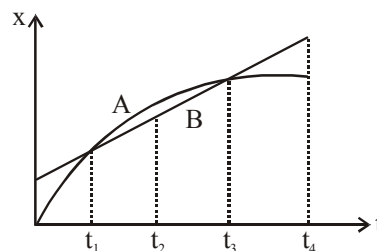
(1) $\frac{mv}{3}$ (2) mv
(3) $2mv$ (4) $mv/2$

28. Five equal resistances each of value R are connected in as shown in figure. The equivalent resistance of the network:-

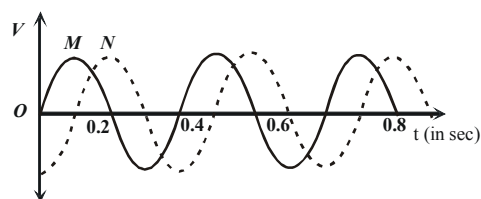


- (1) Between the points B and D is R
 (2) Between the points B and D is $2R$
 (3) Between the points A and C is R
 (4) Between the points A and C is $\frac{R}{2}$
29. A circuit has a resistance of 11Ω , an inductive reactance of 25Ω and a capacitive reactance of 18Ω . It is connected to an ac source of $260V$ and $50Hz$. The current through the circuit (in amperes) is
 (1) 11 (2) 15 (3) 18 (4) 20
30. A compound microscope has an eye piece of focal length 10 cm and an objective of focal length 4 cm . Calculate the magnification, if an object is kept at a distance of 5 cm from the objective so that final image is formed at the least distance vision (20 cm)
 (1) 12 (2) 11
 (3) 10 (4) 13
31. Two thermally insulated vessels 1 and 2 are filled with air at temperatures (T_1, T_2) , volume (V_1, V_2) and pressure (P_1, P_2) respectively. If the valve joining the two vessels is opened, the temperature inside the vessel at equilibrium will be
 (1) $T_1 + T_2$
 (2) $(T_1 + T_2)/2$
 (3) $\frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_2 + P_2 V_2 T_1}$
 (4) $\frac{T_1 T_2 (P_1 V_1 + P_2 V_2)}{P_1 V_1 T_1 + P_2 V_2 T_2}$

32. The graph given shows position of two cars A and B as a function of time. The cars move along x-axis on parallel but separate tracks. At which instant of time is car A overtaking car B ?



- (1) t_1 (2) t_2 (3) t_3 (4) t_4
33. An electric lamp is marked 60 W , 230 V . The cost of 1 kilowatt hour of power is Rs. 1.25. The cost of using this lamp for 8 hours is
 (1) Rs. 1.20 (2) Rs. 4.00
 (3) Rs. 0.25 (4) Rs. 0.60
34. Two sinusoidal voltages of the same frequency are shown in the diagram. What is the frequency, and the phase relationship between the voltages



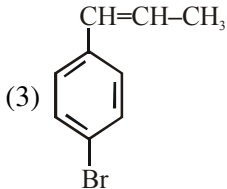
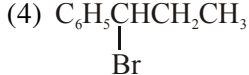
Frequency in Hz Phase lead of N over M in radians
 (1) 0.4, $-\pi/4$ (2) 2.5, $-\pi/2$
 (3) 2.5, $+\pi/2$ (4) 2.5, $-\pi/4$

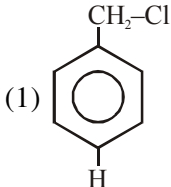
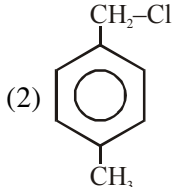
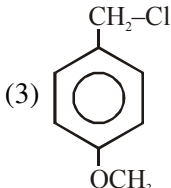
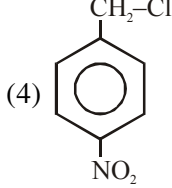
35. If a thin prism of glass is dipped into water then minimum deviation (with respect to air) of light produced by prism will be left $\left({}^a\mu_g = \frac{3}{2} \text{ and } {}^a\mu_w = \frac{4}{3} \right)$
 (1) $\frac{1}{2}$ (2) $\frac{1}{4}$
 (3) 2 (4) $\frac{1}{5}$

- 36.** Two rods of same length and cross section are joined along the length. Thermal conductivities of first and second rod are K_1 and K_2 . The temperature of the free ends of the first and second rods are maintained at θ_1 and θ_2 respectively. The temperature of the common junction is
- (1) $\frac{\theta_1 + \theta_2}{2}$ (2) $\frac{K_2 K_2}{K_1 + K_2}(\theta_1 + \theta_2)$
- (3) $\frac{K_1 \theta_1 + K_2 \theta_2}{K_1 + K_2}$ (4) $\frac{K_2 \theta_1 + K_1 \theta_2}{K_1 + K_2}$
- 37.** A stone, dropped from the top of a tower, travels 35 m in the last second of its journey. The height of the tower is : ($g = 10 \text{ m/s}^2$)
- (1) 20 m (2) 40 m
(3) 60 m (4) 80 m
- 38.** Flux ϕ (in weber) in a closed circuit of resistance 10 ohm varies with time t (in sec) according to the equation $\phi = 6t^2 - 5t + 1$. What is the magnitude of the induced current at $t = 0.25 \text{ s}$?
- (1) 1.2 A (2) 0.8 A
(3) 0.6 S (4) 0.2 A
- 39.** Photons of energy 6 eV are incident on a metal surface whose work function is 4 eV. The minimum kinetic energy of the emitted photo-electrons will be
- (1) 0 eV (2) 1 eV
(3) 2 eV (4) 10 eV
- 40.** In Young's experiment, light of wavelength $4000 \text{ \AA}$ is used to produce bright fringes of width 0.6 mm, at a distance of 2 meters. If the whole apparatus is dipped in a liquid of refractive index 1.5, then fringe width will be
- (1) 0.2 mm (2) 0.3 mm
(3) 0.4 mm (4) 1.2 mm
- 41.** If at temperature $T_1 = 1000\text{K}$, the wavelength corresponding to maximum radiation is $1.4 \times 10^{-6} \text{ m}$, then at what temperature that wavelength will be $2.8 \times 10^{-6} \text{ m}$
- (1) 2000K (2) 500K
(3) 250K (4) None of these
- 42.** A physical quantity A is related to four observations a, b, c and d as follows, $A = \frac{a^2 b^3}{c \sqrt{d}}$. The percentage errors of measurement in a, b, c and d are 1%, 3%, 2% and 2% respectively. What is the percentage error in the quantity A
- (1) 12% (2) 7%
(3) 5% (4) 14%
- 43.** Amplitude of electric field in EM wave is give as $E_0 = 300 \text{ V/m}$. Amplitude of magnetic field will be :-
- (1) 10^{-8} T (2) 10^{-6} T
(3) 10^{-10} T (4) 10^{-7} T
- 44.** In a fission reaction ${}^{236}_{92}\text{U} \rightarrow {}^{117}\text{X} + {}^{117}\text{Y} + n + n$, the binding energy per nucleon of X and Y is 8.5 MeV whereas of ${}^{236}\text{U}$ is 7.6 MeV. The total energy liberated will be approximately
- (1) 200 KeV
(2) 2 MeV
(3) 200 MeV
(4) 2000 MeV
- 45.** A parallel beam of monochromatic light of wavelength $5000 \text{ \AA}$ is incident normally on a single narrow slit of width 0.001 mm. The light is focused by a convex lens on a screen placed on the focal plane. The first minimum will be formed for the angle of diffraction equal to
- (1) 0° (2) 15°
(3) 30° (4) 60°

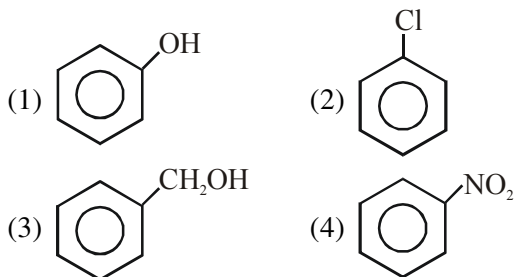
46. The correct bond order in the following species is:-
 (1) $O_2^{2+} < O_2^- < O_2^+$ (2) $O_2^+ < O_2^- < O_2^{2+}$
 (3) $O_2^- < O_2^+ < O_2^{2+}$ (4) $O_2^{2+} < O_2^+ < O_2^-$
47. Which does not form double salt ?
 (1) Li_2SO_4 (2) Na_2SO_4 (3) K_2SO_4 (4) Rb_2SO_4
48. A compound contains 1.2×10^{25} atoms of A, 4.8×10^{25} atoms of B and 2.4×10^{25} atoms of C. Find the empirical formula of compound :-
 (1) ABC_2 (2) ABC_4 (3) AB_4C (4) AB_4C_2
49. In the formation of sulphur trioxide by the contact process $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$; the rate of reaction is expressed as

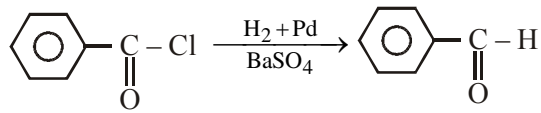
$$-\frac{d[O_2]}{dt} = 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

 The rate of disappearance of (SO_2) will be :-
 (1) $5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (2) $-2.25 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (3) $3.75 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (4) $50.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
50. The reaction of $C_6H_5CH=CHCH_3$ with HBr produces:-
 (1) $C_6H_5CH_2CH(Br)CH_3$ (2) $C_6H_5CH_2CH_2CH_2Br$
 (3) $C_6H_5CHCH_2CH_3$
 (4) $C_6H_5CHCH_2CH_3$
51. Which one of the following molecular hydrides acts as a Lewis acid ?
 (1) CH_4 (2) NH_3 (3) H_2O (4) B_2H_6
52. Lithium chloride is highly soluble in -
 (1) C_6H_6 (2) H_2O (3) D_2O (4) All
53. If $pOH \geq 5$ then possible nature of aqueous solution is :-
 (1) acidic (2) basic
 (3) neutral (4) any one is possible

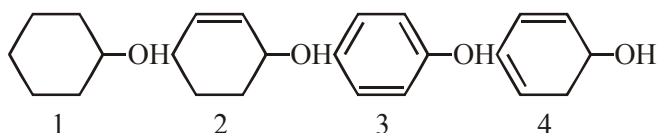
54. A metal X crystallises in a face-centred cubic arrangement with the edge length 862 pm. What is the shortest separation of any two nuclei of the atom?
 $(\sqrt{2} = 1.4)$
 (1) 406 pm (2) 707 pm (3) 862 pm (4) 616 pm
55. Which of the following is most reactive towards S_N1 .
 (1)  (2)  (3)  (4)
56. The basic structural unit of silicates is :-
 (1) SiO_4^{2-} (2) SiO^- (3) SiO_4^{4-} (4) SiO_3^{2-}
57. The metal used in photoelectric cell is :-
 (1) Na (2) K (3) Mg (4) Ca
58. K_a for cyano acetic acid is 4×10^{-4} . Then the degree of hydrolysis of 0.04M sodium cyano acetate solution will be :-
 (1) 2.5×10^{-8} (2) 0.25×10^{-4}
 (3) 25×10^{-8} (4) 0.25×10^{-7}
59. A solution containing 12.5 g of non-electrolyte substance in 250 g of water shows boiling point elevation of 0.52 K. Calculate the molar mass of the substance. ($K_b = 0.52 \text{ K kg mol}^{-1}$)
 (1) 43.06 g mol^{-1} (2) 25.3 g mol^{-1}
 (3) 16.08 g mol^{-1} (4) 50 g mol^{-1}
60. Glucose and fructose are of each other
 (1) homologous
 (2) functional group isomers
 (3) mirror image isomers
 (4) nonisomers
61. Which of the following reaction is not feasible:-
 (1) $F_2 + 2Cl^- \longrightarrow 2F^- + Cl_2$
 (2) $Cl_2 + 2Br^- \longrightarrow 2Cl^- + Br_2$
 (3) $Br_2 + 2F^- \longrightarrow 2Br^- + F_2$
 (4) $Br_2 + 2I^- \longrightarrow 2Br^- + I_2$

62. Minimum lone pair on central atom will be present in:-
 (1) XeO_3 (2) XeF_4 (3) ClF_3 (4) HOCl
63. The solubility of AgI in NaI solution is less than that in pure water because :-
 (1) AgI forms complex with NaI
 (2) of common ion effect
 (3) solubility product of AgI is less than that of NaI
 (4) the temperature of the solution decreases.
64. When does a real gas show behaviour same as ideal gas :
 (1) At low temperature and low pressure
 (2) At high temperature and high pressure
 (3) At low temperature and high pressure
 (4) At high temperature and low pressure
65. Which of the following is most reactive for nucleophilic substitution reaction ?
 (1) CH_3CONH_2 (2) $\text{CH}_3\text{COOC}_2\text{H}_5$
 (3) $\text{CH}_3\text{C}(=\text{O})\text{O}\text{C}(=\text{O})\text{CH}_3$ (4) $\text{CH}_3\text{C}(=\text{O})\text{Cl}$
66. Aluminium is purified by :-
 (1) Roasting
 (2) Sublimation
 (3) Electrolytic refining
 (4) Reduction with carbon
67. Dative Bond is present in :-
 (1) KI_3 (2) KNO_2 (3) K_2CO_3 (4) All
68. For which of the following reactions is product formation favoured by low pressure and high temperature ?
 (1) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g}) \Delta H^\circ = -9.4 \text{ kJ}$
 (2) $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightleftharpoons 2\text{CO}(\text{g}) \Delta H^\circ = +172.5 \text{ kJ}$
 (3) $\text{CO}(\text{g}) + 2\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{l}) \Delta H^\circ = -21.7 \text{ kJ}$
 (4) $3\text{O}_2(\text{g}) \rightleftharpoons 2\text{O}_3(\text{g}) \Delta H^\circ = +285 \text{ kJ}$
69. Which one of the following is most reactive towards electrophilic substitution reaction ?



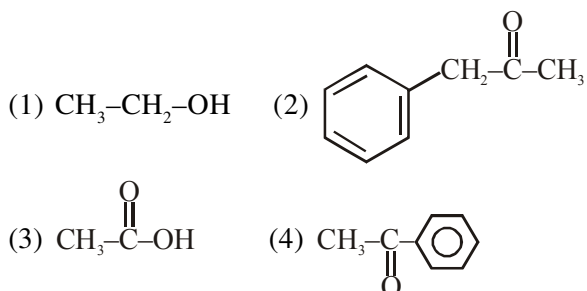
70. Polymer which has amide linkage is
 (1) Nylon-6,6 (2) Terylene
 (3) Teflon (4) Bakelite
71. Argentite is an ore of :-
 (1) Iron (2) Gold
 (3) Silver (4) Platinum
72. A molecule XY_2 contains two σ , two π bond and one lone pair of electron in the valence shell of X. The arrangement of lone pair as well as bond pair is :
 (1) Square pyramidal (2) Linear
 (3) Trigonal planar (4) Unpredictable
73. 200 J of heat is provided to the system. Work done on the system is 20J. What is the change in internal energy?
 (1) 220 kJ (2) $2.2 \times 10^{-1} \text{ kJ}$
 (3) $2.2 \times 10^{-5} \text{ kJ}$ (4) 220 cal
74. When huge amount of sewage is dumped into a river, the BOD will –
 (1) Increase
 (2) Remain unchanged
 (3) Slightly decrease
 (4) Decrease
75. The given reaction is

 (1) Mendius reaction
 (2) Stephen's reaction
 (3) Rosenmund's reduction
 (4) Cannizzaro's reaction
76. Which of the following has least conductivity in aqueous solution.
 (1) $\text{Co}(\text{NH}_3)_4\text{Cl}_3$ (2) $\text{Co}(\text{NH}_3)_3\text{Cl}_3$
 (3) $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ (4) $\text{Co}(\text{NH}_3)_6\text{Cl}_3$
77. Methylene blue dye becomes colourless if mixed with :-
 (1) H_2O (2) CuSO_4
 (3) $\text{Cr}(\text{OH})_3$ (4) Charcoal
78. CsBr has b.c.c. structure with edge length 4 Å. The shortest inter ionic distance between Cs^+ and Br^- is:
 (1) 3.46 Å (2) 1.86 Å
 (3) 7.44 Å (4) 4.3 Å

79. Dehydration of following alcohols will be in order:-
(by conc. H_2SO_4)



- (1) $1 < 2 < 3 < 4$ (2) $4 > 3 > 1 > 2$
(3) $4 > 2 > 1 > 3$ (4) $1 > 3 > 4 > 2$

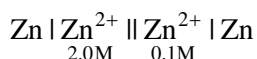
80. Which of the following compound will not give Iodoform Test :-



81. Which gives only one mole of AgCl , when reacts with AgNO_3 :-

- (1) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (2) $\text{PtCl}_4 \cdot 5\text{NH}_3$
(3) $\text{PtCl}_4 \cdot 2\text{NH}_3$ (4) $\text{PtCl}_4 \cdot 3\text{NH}_3$

82. In the electrochemical cell

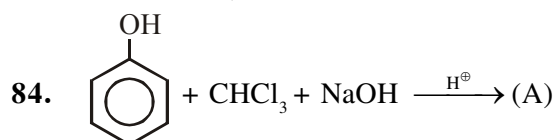


Calculate emf of the cell :- ($\log 20 = 1.3$)

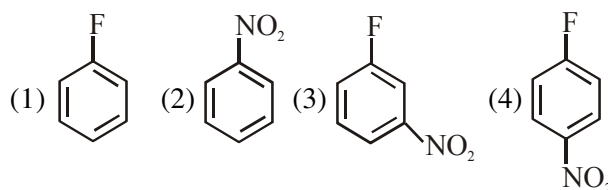
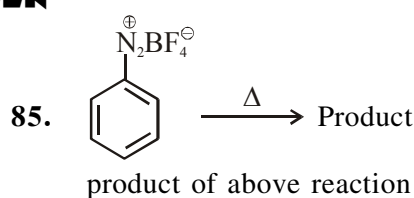
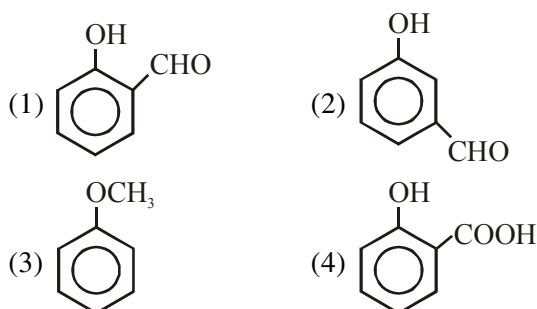
- (1) +78 mV (2) -78 mV
(3) +39 mV (4) -39 mV

83. Which of following is reductant :

- (1) $\text{K}_2\text{Cr}_2\text{O}_7$ (2) KMnO_4 (3) H_2SO_4 (4) H_2S



Product (A) is :-



86. The outer electronic configuration of Gadolinium (at no.-64) is :-

- (1) $4f^7 5d^1 6s^2$ (2) $4f^8 5d^0 6s^2$
(3) $4f^9 5d^0 6s^2$ (4) $4f^7 5d^0 6s^2$

87. Electrolysis of aqueous AgNO_3 causes :-

- (1) Decrease in pH (2) Increase in pH
(3) Constant pH (4) None of these

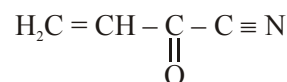
88. If x is the fraction of molecules having energy greater than E_a , it will be given by :-

- (1) $x = -\frac{E_a}{RT}$ (2) $\ln x = -\frac{E_a}{RT}$
(3) $x = e^{E_a/RT}$ (4) None of these

89. In the reaction  $\xrightarrow{\text{HBr}}$ the products are

- (1)  and H_2
(2)  and CH_3Br
(3)  and CH_3OH
(4)  and CH_3Br

90. The no. of σ and π bonds are present in given compound.

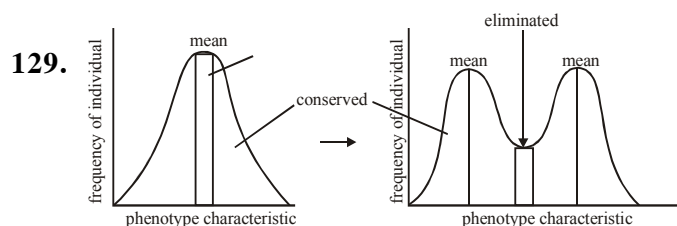


- (1) 8 σ and 4 π (2) 10 σ and 5 π
(3) 8 σ and 3 π (4) 9 σ and 4 π

- | | |
|--|--|
| <p>91. Which one cell structure is found in both prokaryotic and Eukaryotic cells?
 (1) Plastids (2) Mitochondria
 (3) Ribosomes (4) Golgi body</p> <p>92. Ion that stimulates entry of Ca^{+2} in pre-synaptic neuron:-
 (1) K^{+} (2) Ca^{+2} (3) Na^{+} (4) Cl^{-}</p> <p>93. Which of the following is not a lateral meristem?
 (1) Phellogen
 (2) Vascular cambium
 (3) Intra fascicular cambium
 (4) Intercalary meristem</p> <p>94. Trypsin is :-
 (1) An alkaloid (2) An enzyme
 (3) A drug (4) A toxin</p> <p>95. Organ system level of organization is observed in:-
 (1) Annelida (2) Mollusca
 (3) Mammalia (4) All of the above</p> <p>96. Which one is not a detrimental interaction ?
 (1) Predation (2) Amensalism
 (3) Competition (4) Commensalism</p> <p>97. Which of the following is not a characteristic feature of genetic code:-
 (1) Mostly universal (2) Comma less
 (3) Overlapping (4) Triplet</p> <p>98. Components of blood, which play role in transport of gases:-
 (1) Plasma and RBC
 (2) Plasma and WBC
 (3) Plasma and Platelets
 (4) RBC and WBC</p> <p>99. Group of neurosecretory cells are present in ?
 (1) Thalamus (2) Anterior pituitary
 (3) Hypothalamus (4) Cerebrum</p> <p>100. Surgical removal of uterus is :-
 (1) Tubectomy (2) Hysterectomy
 (3) Vasectomy (4) Orchidectomy</p> <p>101. The main lipid component of cell membrane is
 (1) Cholesterol (2) Glycolipid
 (3) Palmitic acid (4) Phospholipid</p> | <p>102. Pons varoli is characterised by a centre called :-
 (1) Hunger centre
 (2) Thirst centre
 (3) Cardiac acceleratory centre
 (4) Pneumotaxic centre</p> <p>103. Which of the following is not included in periderm?
 (1) Phellem (2) Phellogen
 (3) Phelloderm (4) Secondary phloem</p> <p>104. Which one of the following nitrogenous base is absent in DNA :-
 (1) Adenine (2) Guanine
 (3) Uracil (4) Thymine</p> <p>105. Water vascular system is the feature of which of the following phylum of animal kingdom ?
 (1) Annelida (2) Arthropoda
 (3) Mollusca (4) Echinodermata</p> <p>106. Reason of species extinction in natural ecosystem according to Gause principle :-
 (1) Competition (2) Predation
 (3) Co-extinction (4) Resources partitioning</p> <p>107. Which of the following RNA provides template for protein synthesis-
 (1) m-RNA (2) t-RNA
 (3) r-RNA (4) Si-RNA</p> <p>108. Genetic material of HIV is ?
 (1) ds-RNA (2) ds-DNA
 (3) ss-RNA (4) ss-DNA</p> <p>109. Hisardale is an example of :-
 (1) Inbreeding depression
 (2) Out-crossing
 (3) Cross-breeding
 (4) Interspecific hybridisation</p> <p>110. Secondary sexual characters in female develop in response to hormone :-
 (1) Relaxin (2) Progesterone
 (3) Estrogen (4) Gonadotropin</p> <p>111. The core of cilia and flagella is called -
 (1) Radial spoke (2) Axoneme
 (3) Central hub (4) Central sheath</p> |
|--|--|

112. Nissl's granules are found in _____ and helps in _____.
 (1) Cyton only, Lipid synthesis
 (2) Cyton & Axon, Protein synthesis
 (3) Dendron & Axon, Protein synthesis
 (4) Dendron & Cyton, Protein synthesis
113. Which of the following is a part of ground tissue system in the leaves of flowering plants ?
 (1) Medullary rays (2) Pericycle
 (3) Pith (4) Mesophyll
114. The water potential of pure water at standard temperatures, which is not under any pressure, is taken to be :-
 (1) Zero (2) 1
 (3) 10 (4) 100
115. Longest and strongest bone of our body is :-
 (1) Scapula (2) Tibia
 (3) Femur (4) Fibula
116. Which one structural feature of community ?
 (1) Energy flow (2) Decomposition
 (3) Stratification (4) Mineral cycle
117. Inheritance of flower colour in *Antirrhinum majos* shows:-
 (1) Dominance (2) Codominance
 (3) Incomplete dominance (4) Linkage
118. Which of the following diagnostic test is done for Mary Mallon disease ?
 (1) Biopsy (2) Schick test
 (3) Widal Test (4) MRI
119. Apiculture is related to :-
 (1) *Bombyx mori* (2) *Apis indica*
 (3) *Locusta* (4) Lac insect
120. In uterus, endometrium proliferates in response to
 (1) Relaxin (2) Oxytocin
 (3) Progesterone (4) Oestrogen
121. Which one cell organelle is not a part of Endo-membrane system?
 (1) Lysosome
 (2) Golgi body
 (3) Endoplasmic reticulum
 (4) Peroxisome

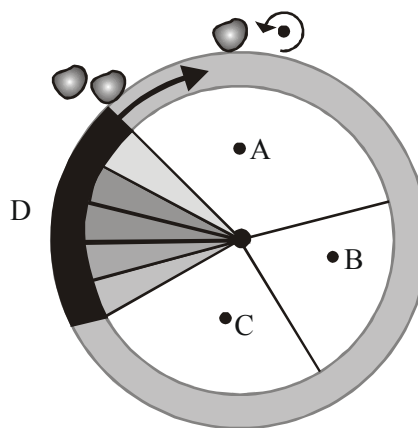
122. Kinetic balance is related with
 (1) Organ of Corti (2) Vestibular apparatus
 (3) Ampulla (4) Eustachian tube
123. Basal part of the ovule is :-
 (1) Micropyle (2) Chalaza
 (3) Hilum (4) Funicle
124. How many ATP are required in biological nitrogen fixation for the formation of 2 NH₃ ?
 (1) 8 ATP (2) 32 ATP
 (3) 16 ATP (4) 4 ATP
125. Cardiac muscles are the muscles of _____ :-
 (1) Stomach (2) Heart
 (3) Intestine (4) Kidney
126. Interaction of biotic factors with each other and they integrated with its physical environment creates :-
 (1) Community (2) Ecosystem
 (3) Population (4) Organism
127. Parbhani Kranti, which has resistance to yellow mosaic virus is a variety of
 (1) Wheat (2) Bhindi (3) Rice (4) Bajra
128. Which of the following is not a bacterial disease?
 (1) Typhoid (2) Diphtheria
 (3) Pneumonia (4) Amoebiasis



129. This graph indicates which type of natural selection in a population :-
 (1) Stabilizing selection (2) Disruptive selection
 (3) Direction selection (4) Mutation
130. Pregnancy begins with implantation of
 (1) Morula (2) Fertilized ovum
 (3) Blastopore (4) Blastocyst
131. Which one structure is not found in nucleoplasm?
 (1) Centrioles (2) Chromatin fibre
 (3) r-RNA (4) Nucleolus
132. Cell wall of bacteria is made up of :-
 (1) Peptidoglycan (2) Cellulose
 (3) Chitin (4) Starch

- 133.** *Hydrilla* is pollinated by :-
 (1) Air (2) Water
 (3) Insect (4) Bat
- 134.** Law of limiting factors was proposed by :-
 (1) Joseph Priestley (2) Jan Ingenhousz
 (3) Cornelius van Niel (4) F. F. Blackman
- 135.** In cockroach, abdomen in both males and females consists of _____ segments :-
 (1) 8 (2) 9 (3) 10 (4) 12
- 136.** Lipophilic toxic substances increase at successive trophic level leads to :-
 (1) Biomagnification
 (2) Eutrophication
 (3) Cultural eutrophication
 (4) BOD
- 137.** Which of the following is "Clot buster"–
 (1) Streptokinase (2) Cyclosporin-A
 (3) Statin (4) Citric acid
- 138.** LSD is a powerful drug obtained from ?
 (1) *Erythroxylum Coca* (2) *Cannabis Sativa*
 (3) *Claviceps purpuria* (4) *Papaver somniferum*
- 139.** Phylogeny term represent :-
 (1) Life history (2) Group of phyla
 (3) Evolutionary history (4) Group of species
- 140.** Oral contraceptives are used by females to check
 (1) Implantation
 (2) Fertilisation
 (3) Ovulation
 (4) Entry of sperms in vagina
- 141.** In which phase of cell cycle spindle fibres attach to kinetochore of chromosome?
 (1) Prophase (2) Metaphase
 (3) Anaphase (4) Telophase
- 142.** Stored food of red algae is :-
 (1) Floridean starch (2) Chitin
 (3) Laminarin (4) Glycogen
- 143.** In *Dianthus*, placentation is :-
 (1) Parietal (2) Axile
 (3) Free central (4) Basal
- 144.** Respiratory quotient (R.Q.) or respiratory ratio is:-
 (1) O_2/CO_2 (2) CO_2/O_2
 (3) H_2O/CO_2 (4) CO_2/H_2O
- 145.** Primary producer → A
 ↓
 Primary consumer → Rabbit
 ↓
 Secondary consumer → B
 ↓
 Tertiary consumer → C
 (1) A-Grass, B-Wolf, C-Lion
 (2) A-Trees, B-Goat, C-Birds
 (3) A-Phytoplanktons, B-Small fish, C-Birds
 (4) A-Wolf, B-Lion, C-Trees
- 146.** The government of India has passed the water (prevent and control of pollution) ACT in :-
 (1) 1974 (2) 1981 (3) 1987 (4) 1990
- 147.** Transgenic animal produces biological product. Such as α -1 antitrypsin, which is used to treat–
 (1) Sick cell anaemia (2) Haemophilia
 (3) Emphysema (4) Colour blindness
- 148.** Cancer of lymph nodes and other similar tissue derived from primitive mesoderm are very rare in human. Such type of Cancer fall under the category of ?
 (1) Carcinoma (2) Sarcoma
 (3) Melanoma (4) Leukaemia
- 149.** Which period is also called as 'Golden age of dinosaurs' :-
 (1) Triassic (2) Cretaceous
 (3) Jurassic (4) Azoic
- 150.** Which of the following is the most important cause of animals and plants being driven to extinction?
 (1) Over - exploitation
 (2) Alien species invasion
 (3) Habitat loss and fragmentation
 (4) Co-extinctions
- 151.** Tetrad of homologous chromosomes is clearly visible in
 (1) Leptotene (2) Zygotene
 (3) Pachytene (4) Diplotene
- 152.** Loose smut disease of wheat is caused by :-
 (1) *Ustilago* (2) *Puccinia*
 (3) *Albugo* (4) *Rhizopus*
- 153.** Polyadelphous condition is present in :-
 (1) Citrus (2) Cotton (3) Pea (4) Chinrose

- 154.** Which of the following plant growth regulators promotes bolting in plants ?
 (1) Auxin (2) Cytokinin
 (3) Absciscic acid (4) Gibberellin
- 155.** Humus is degraded by some microbes and release of inorganic nutrients occur by the process known as :-
 (1) Decomposition (2) Humification
 (3) Mineralisation (4) Catabolism
- 156.** Mineral which play role in blood clotting:-
 (1) Ca^{+2} (2) Mg^{+2} (3) K^{+} (4) Na^{+}
- 157.** Golden rice variety rich in-
 (1) Vitamin-A (2) Fatty acid
 (3) Vitamin-C (4) Cholesterol
- 158.** If a person suffering from fever due to any infection. Which type of immunity will developed in his body.
 (1) Active, artificial (2) Passive, artificial
 (3) Active, natural (4) Passive, natural
- 159.** The process of evolution is :-
 (1) Continuous
 (2) Discontinuous
 (3) Continuous in the begining but now discontinuous
 (4) Discontinuous in the begining but now continuous
- 160.** The term ecosystem was coined by :-
 (1) E.P. Odum (2) A.G. Tansley
 (3) E. Haeckel (4) E. Warming
- 161.** Which one phase of cell division is helpful for increases the genetic variability in the population of organisms from one generation to next?
 (1) Metaphase (2) Prophase-I
 (3) Telophase (4) Anaphase-I
- 162.** Citrus canker disease is caused by :-
 (1) Fungi (2) Bacteria
 (3) Mycoplasma (4) Virus
- 163.** Floral formula of pea is :-
 (1) $\oplus \overline{\text{K}}_{(5)} \text{C}_{(5)} \text{A}_5 \underline{\text{G}}_{(2)}$ (2) $\% \overline{\text{K}}_{(5)} \widehat{\text{C}}_{(5)} \text{A}_{(9)+1} \underline{\text{G}}_{(2)}$
 (3) $\% \overline{\text{K}}_{(5)} \text{C}_{1+2+(2)} \text{A}_{(9)+1} \underline{\text{G}}_1$ (4) $\oplus \overline{\text{K}}_{(5)} \text{C}_{1+(2)+(2)} \text{A}_{10} \underline{\text{G}}_1$
- 164.** Enzymes, catalysing a transfer of any group (other than hydrogen) between substrates, are classified under class :-
 (1) Transferases (2) Oxidoreductases
 (3) Hydrolases (4) Lyases
- 165.** A traditional strategy for the protection of biodiversity in India is :-
 (1) Ex-situ conservation (2) In-situ conservation
 (3) Biosphere reserve (4) Sacred forest and lake
- 166.** Which of the following heart chamber has thickest wall:-
 (1) Right Atrium (2) Right ventricle
 (3) Left Atrium (4) Left ventricle
- 167.** RNAi is used for :-
 (1) Production of insulin
 (2) Production of ethylene
 (3) Production of Nematode resistant plant
 (4) Production of statin
- 168.** During time of stress body demands the releasing of cortisol and adrenaline both having receptor respectively present on :-
 (1) Intracellular both
 (2) Cell membrane, Intracellular
 (3) Intracellular, Cell membrane
 (4) Cell membrane both
- 169.** "Geological Time scale was firstly proposed by :-
 (1) Buffon (2) A.R. Wallace
 (3) Giovanni avduina (4) Darwin
- 170.** Which of the following would appear as the pioneer organisms on bare rocks?
 (1) Lichens (2) Liverworts
 (3) Mosses (4) Green algae
- 171.** In the given figure of cell cycle choose the phase (A to D) in which DNA replication takes place ?



- (1) A (2) B (3) C (4) D

- 172.** Which type of vascular bundles are found in maize root ?
 (1) Conjoint, collateral and closed
 (2) Conjoint, collateral and open
 (3) Leptocentric
 (4) Radial
- 173.** Twisted aestivation is present in petals of :-
 (1) *Calotropis* (2) Cotton
 (3) Pea (4) Cassia
- 174.** Lining of body cavities, ducts and tubes are made up of :-
 (1) Transitional epithelium
 (2) Compound epithelium
 (3) Keratinised epithelium
 (4) Simple epithelium
- 175.** Which one factor is not include in evil quartet :-
 (i) Habitat loss and fragmentation.
 (ii) Over exploitation
 (iii) Alien species invasion
 (iv) Pollution and disturbance
 (v) Co-extinction
 (1) v (2) iv
 (3) ii (4) iii
- 176.** How much amount of urea exerted by human perday normally :-
 (1) 25–30mg (2) 4mg
 (3) 25–30 grams (4) 4 grams
- 177.** Dental formula of adult human is :-
 (1) $\frac{2102}{2102}$ (2) $\frac{2123}{2123}$ (3) $\frac{2132}{2132}$ (4) $\frac{2123}{2132}$
- 178.** Which of the following hormone is not secreted by adenohypophysis ?
 (1) ACTH (2) ADH
 (3) TSH (4) GH
- 179.** 'Eye of octopus' and 'eyes of mammals' are example of :-
 (1) Homologous organ
 (2) Analogous organ
 (3) Vestigial organ
 (4) Atavisin
- 180.** Which is an ex-situ conservation ?
 (1) Seed bank
 (2) National park
 (3) Biosphere reserve
 (4) Sanctuary



Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

Note : In case of any Correction in the test paper, please mail to dlpcorrections@allen.ac.in within 2 days along with Paper code and Your Form No.

नोट: यदि इस प्रश्न पत्र में कोई Correction हो तो कृपया Paper code एवं आपके Form No. के साथ 2 दिन के अन्दर dlpcorrections@allen.ac.in पर mail करें।

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